

Formula List

Functions

Laws of Exponents

$$a^0 = 1$$

$$a^n \times a^m = a^{n+m}$$

$$(a^n)^m = a^{n \times m}$$

$$\frac{a^n}{a^m} = a^{n-m}$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$a^{-n} = \frac{1}{a^n}$$

Laws of Logarithmic Functions

$$\log_b 1 = 0, b > 0$$

$$\log_b b = 1, b \neq 0$$

$$\log_b MN = \log_b M + \log_b N$$

$$\log_b \frac{M}{N} = \log_b M - \log_b N$$

$$\log_b N^k = k \log_b N$$

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Recursion

Formulae to Simplify Solutions Obtained by Iteration

$$1 + r + r^2 + r^3 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1} \quad \text{for all integers } n \geq 0$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad \text{for all integers } n \geq 1$$